

MA 220: Partial Differential Equations

Course Code	MA 220
Course Name	Partial Differential Equations
L-T-P-C	3-1-0-4
Prerequisites	Computer Organization and Processor Architecture Design (EE 326) or equivalent.
Intended for	B.Tech Maths and Computing
Distribution	Core for B. Tech Maths and Computing and Elective for other discipline
Mutual Exclusion	MA 522
Approval	59th BoA (Apr 2025)

Course Contents

- **Unit 1:** Recall the definition of directional derivatives, chain rules and total derivatives, Introduction to PDE, First order PDEs, Solution methods for first order PDE. [5 Lectures]
- **Unit 2:** Classification of first order PDE, Solutions of linear and non-linear PDEs, Cauchy Problem, Cauchy Kowalevski Theorem. [5 Lectures]
- **Unit 3:** Classification of Second Order Partial Differential Equations: normal forms and characteristics. Initial and Boundary Value Problems: Green's identity. [7 Lectures]
- **Unit 4:** Methods of Solution, Methods of separation of variables, Characteristic method, Green's function, Fourier transform. [7 Lectures]
- **Unit 5:** Laplace equation: mean value property, weak and strong maximum principle, Green's function, Poisson's formula, Dirichlet's principle. [5 Lectures]
- **Unit 6:** Heat equation: initial value problem, fundamental solution, weak and strong maximum principle and uniqueness results. Application of Heat equation. [5 Lectures]
- **Unit 7:** Wave equation: uniqueness, D'Alembert's method, method of spherical means and Duhamel's principle. Application of Wave equation. [5 Lectures]

Text Books

1. E. C. Zachmanoglou and D. W. Thoe, **Introduction to Partial Differential Equations**, Dover Publications, New York, 2016.
2. L.C. Evans, **Partial Differential Equations**, Graduate Studies in Mathematics, Vol. 19, AMS, Providence, 1998

References:

1. G. B. Folland, **Introduction to Partial Differential Equations**, Princeton University Press, 1995
2. A. K. Nandakumaran and P. S. Datti, **Partial Differential Equations**, Cambridge-IISC Series, 2020.
3. W. E. Schiesser and G. W. Griffiths, **A Compendium of Partial Differential Equation Models**, Cambridge University Press, 2009.